Part 1 general physics

Define:

Acceleration(a=Δv/t):_____unit:_____

Force(F=Δp/t）:_____unit:_____

Impulse(=Ft):_____unit:_____

Density(=m/V):_____unit:_____

Pressure(=F/A) :_____unit:_____

gravitational field strength(g=F/m):_____ _____unit:____

Power(=W/t)_____unit:_____

Moment(=Fd):_____unit:__

Momentum(=mv):_____unit:__

Work(=FS)_____unit:__

State what is meant by:

Hooke’s law (F=kx):_____

_____

principle of the conservation of momentum:_____

$$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$$

_____

note : v is “+”value in positive direction ,but “—”value in negative direction

- Conditions for a body in _____平衡:

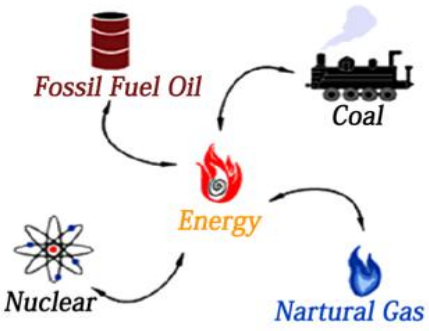
i:_____

ii:_____

- _____再生能源&_____不可再生能源

Non-Renewable Energy

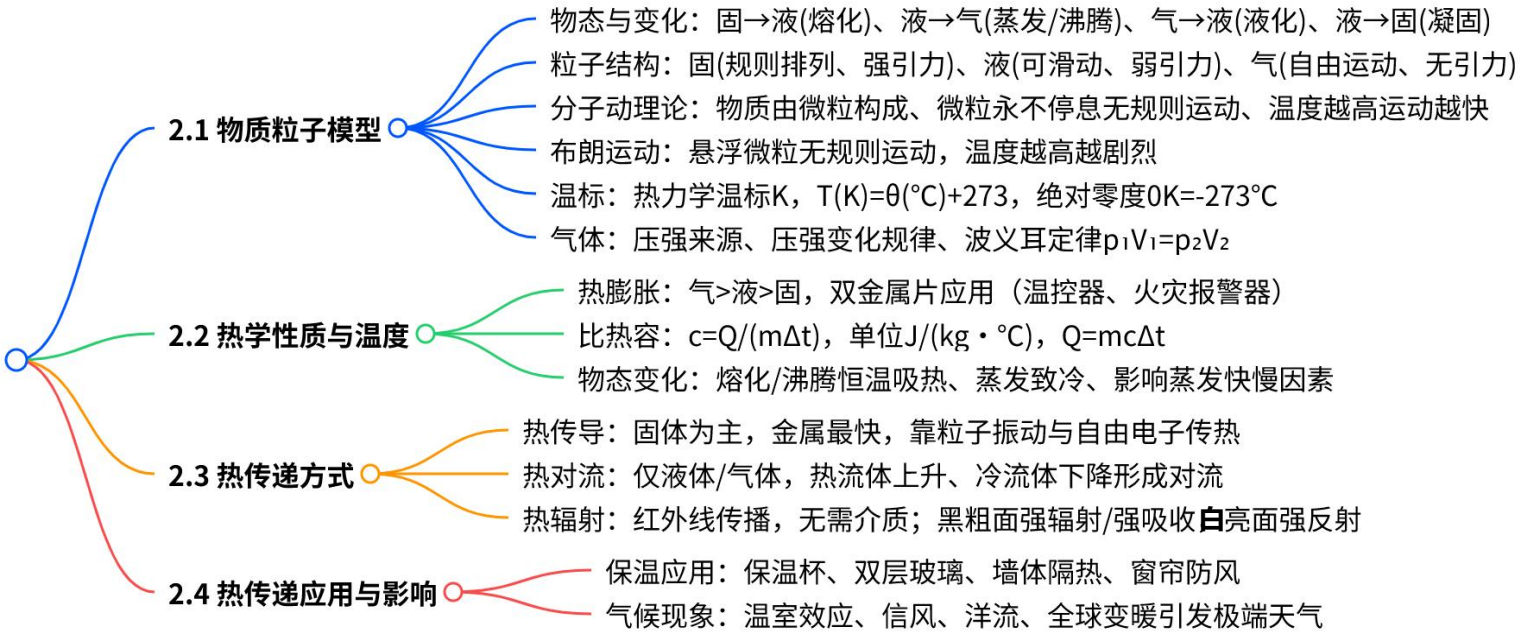
不可再生能源都是燃料类型的，用完就没啦：



the Sun is the source of energy for **all** our energy resources

except 排除_____地热能, _____核能 and _____潮汐能

IGCSE Physics----- Thermal Physics Key Summary



1 Kinetic particle model of matter

(1)the particle structure of solids, liquids and gases

Solid Fixed shape and volume	**Liquid** Fixed shape changed volume	**Gas** no fixed shape and volume
•Strong attractive force hold Particles close together • _____规则排列 • _____振动 in fixed positions	•Weaker attractive forces than solids to hold Particles together. • _____不规则排列 •Particles vibrate but can _____滑动over each other	• no attractive forces between particles •Particles _____ apart to each other • Particles move _____ and _____

(2) evidence for the kinetic molecular model of matter

_____布朗运动:_____

_____

(3) **simple kinetic molecular model of matter物质的分子动理论模型**

i, Matter is made up of very tiny particles (_____分子, _____原子, _____离子**)**

ii, All the particles can move _____随机的 **and** continuously持续的.

iii, The higher the temperature, the _____ **the particles move.**

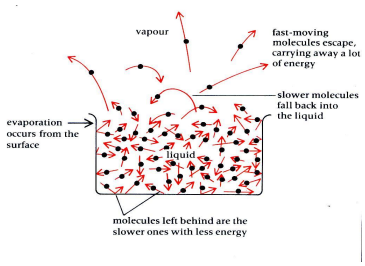
temperature is a measure of _____平均动能 **of molecules.**

- _____**内能--the sum of** _____**分子动能** and _____**分子势能.**

➢ why evaporation makes things cooler?为什么蒸发让物体变冷

_____

_____



➢ **what is the difference between boiling and** _____蒸发:

(i) _____

(ii)_____

（4）the _____开尔文温度（热力学温标） **unit:**_____**(K)**

_____绝对零度:_____the point at which molecules have no kinetic energy

absolute zero=0K=_____**°C**

T (K) = θ (°C) + _____

_____ 温度计

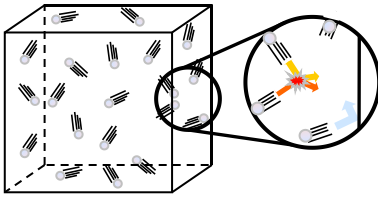
(5) Gases

➢ **Explain gas pressure** in terms of the change of momentum of the particles striking the walls creating a force.

_____

_____

_____



➢ Pressure changes

•**If the volume is constant, then increasing the temperature will increase the pressure.**

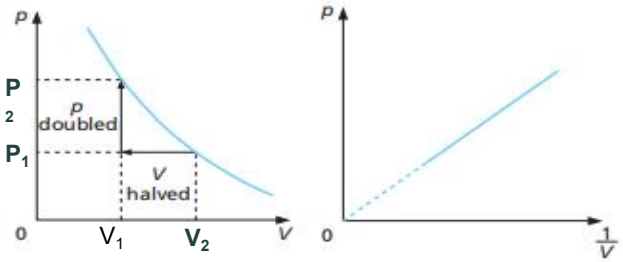
explain:_____.

If the temperature is constant, then decreasing the volume will increase the pressure.

explain:_____

➢ **For a fixed mass of gas, at constant temperature, its pressure is_____反比 to its volume.**

pV = constant or $P_1 \times V_1 = P_2 \times V_2$



2 Thermal properties and temperature

(1) _____**热膨胀** of solids, liquids and gases

_____加热后分子振动更剧烈，间距变大，物体体积变大

expansion is greatest in_____, less in_____ and lowest in_____

(2)_____比热容 $C = \frac{Q}{m\Delta t}$

- define :_____ **unit:** _____

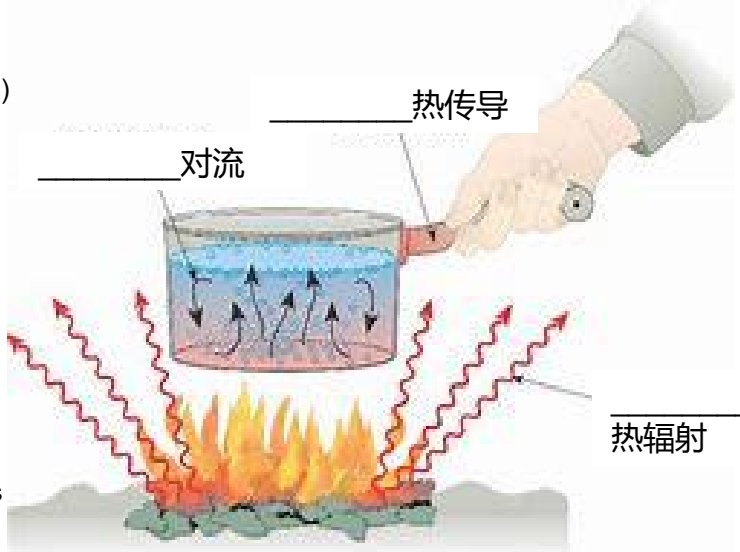
3._____热传递

(1)_____热传导is best in _____（solids，liquids or gas)
energy passed by particle _____振动。

➢ metal is better _____热导体but worse____绝缘体
because energy passed by ion vibration and a better way
---electrons are free to move randomly and collide with atoms
then pass on the energy.

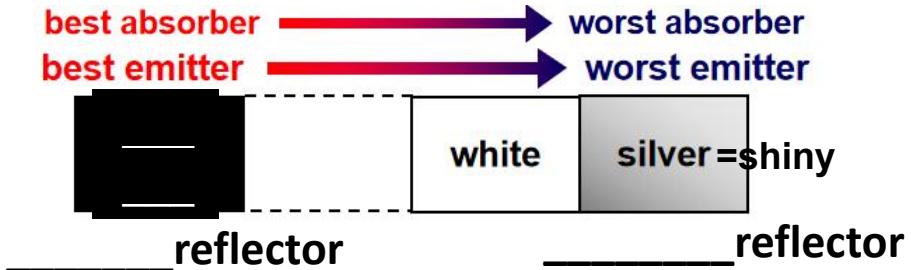
(2)_____对流 occurs in _____and _____
（solids，liquids or gas)

Warm up→ less dense → rises; cool down→ more dense→ sinks



•(3)_____热辐射： All objects **emit** (give out) thermal energy in the form of_____红外线 but very hot objects also give out light waves.

• An emitter will cool down quickly, an absorber will heat up more quickly and a reflector will not heat up quickly



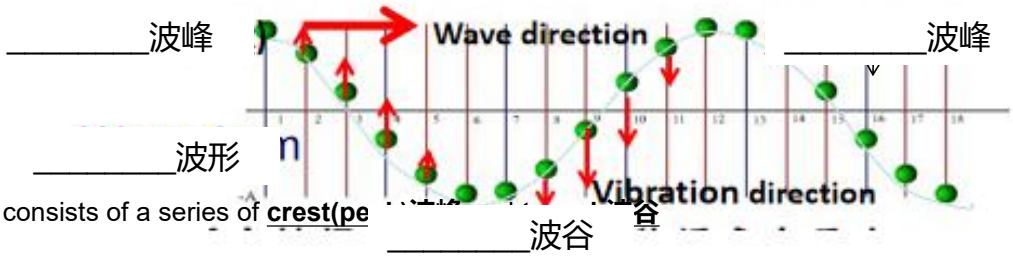
3 waves

1 General properties of waves

(1)two forms:根据传播方向分类

- _____横波:_____

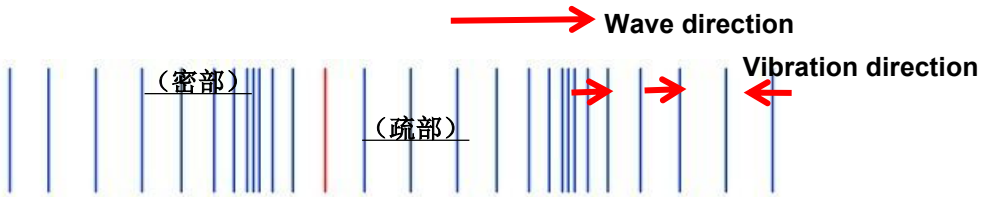
Examples：_____电磁波 ,_____水波



wave form波形 consists of a series of **crest(pe** **谷**

- _____纵波:_____

Examples:_____**声波**



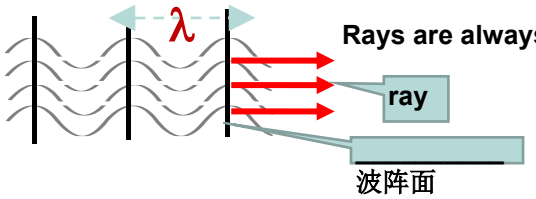
wave form波形 consists of a series of _____密部**(partivle close together)** and _____**疏部(particles furthest away)**

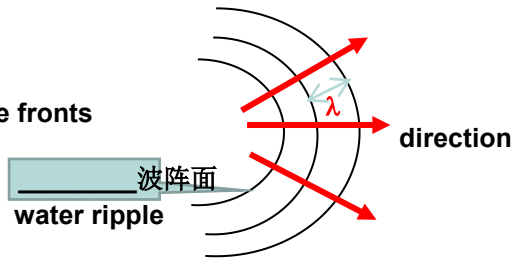
- In an earthquake there are two types of seismic waves: primary P-waves and secondary S-waves.

P-waves are_____These waves are called **primary waves** as they travel_____ and so are felt first.

S- waves are_____**called Secondary wave**

(2) to describe the wave:





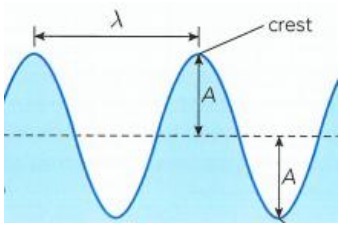
(1)A---amplitude（振幅）:_____

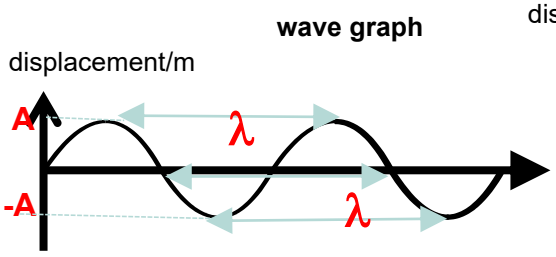
(2)λ----wavelength（波长）：_____unit :m

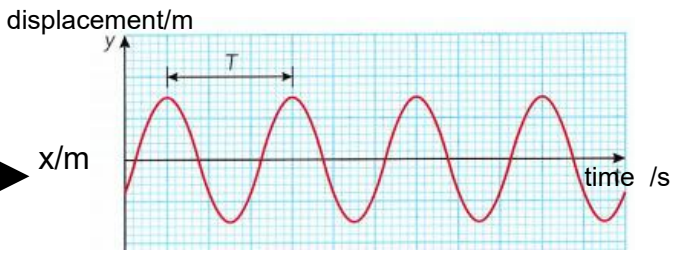
(3)f---frequency（频率）:_____. **f=1/T unit: hertz (Hz赫兹) = 1/second (1/s)**

(4)T---period（周期）：_____. **T=1/f unit:s**

(5)v---wave speed（波速）v=_____公式 **unit: m/s**

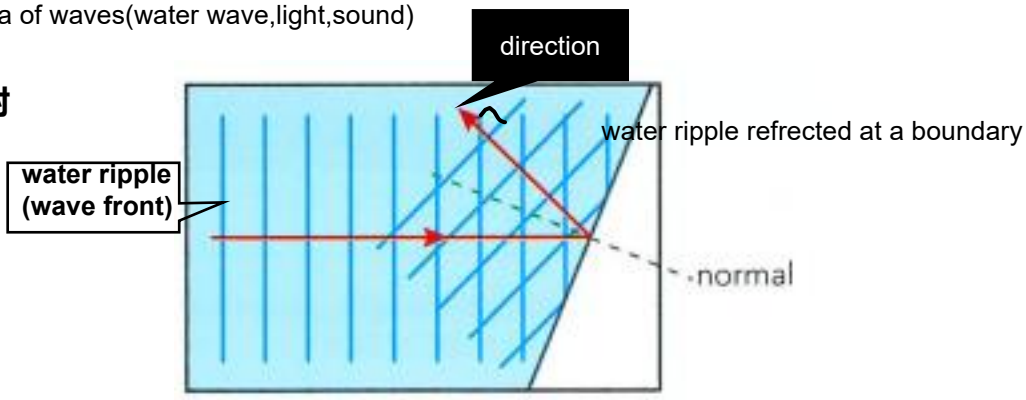






(4) the common phenomena of waves(water wave,light,sound)

➢ _____反射



- Waves bounce away from surface at same angle they strike it.
- **the law of reflection反射定律:** _____入射角）＝_____反射角

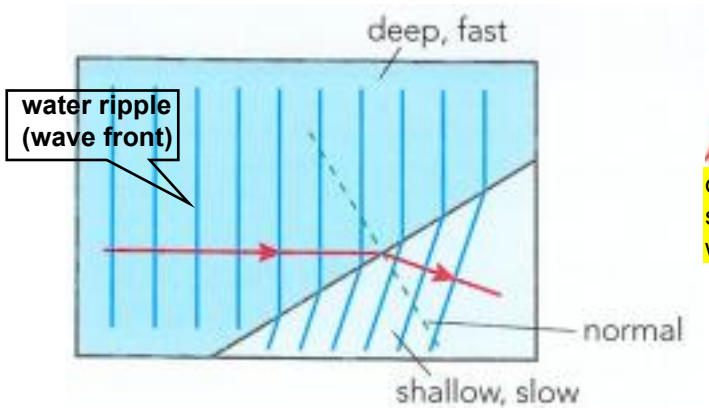
- The plane mirror image像 has these properties:
  - _____
  - _____
  - _____左右颠倒
  - Image is_____虚幻的:

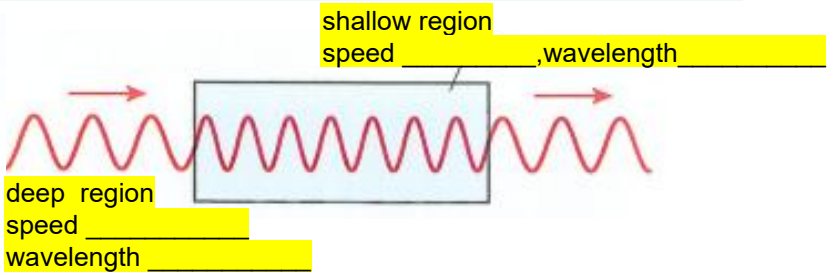
➢ _____折射

_____

_____由于不同介质中的波速不同导致的波的传播方向变化

e.g. water wave move _____慢 in shallow region, but _____快 in deep region,so water wave is refracted when it passing from the deep region to the shallower region.





The speed of light in vacumn(air) is _____, it will _____变慢in some medium(water, glass) .
The more the light is _____, the more it bends _____ the normal line.
the _____ the angle to the normal line.

_____折射率 n = _____ = _____

in medium: n>1
greater n-------> smaller speed---> smaller angle
(means "more refracted=more change in direction")

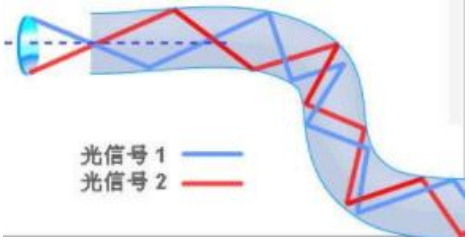
the medium with **greater n (smaller v) is _____ 光密材料**

➢ _____(TIR)全反射

When light passes from an _____ optical dense to a _____ dense medium, all the incident light is _____ inside the denser medium

Critical angle C（临界角）: _____

$$n = \frac{\sin 90^0}{\sin C} = \frac{1}{\sin C} \text{---} > \sin C = _____$$



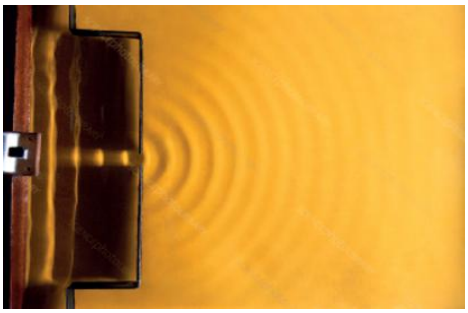
➢ _____光导纤维 ---the use of total internal reflection

➢ _____衍射

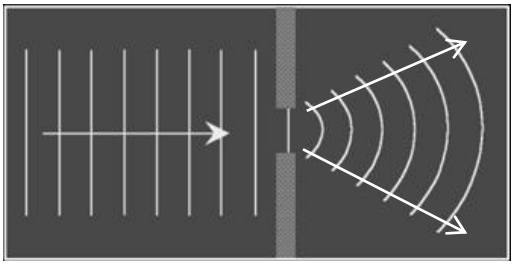
the _____ of waves when they pass through an 狭缝 or round an _____

波长越长衍射现象越容易

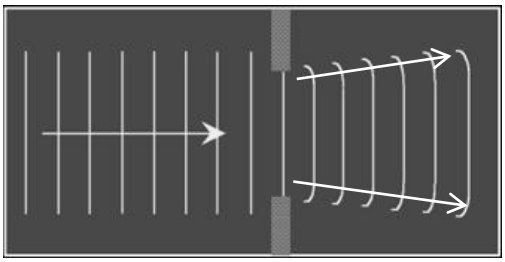
Diffraction water waves in a ripple tank



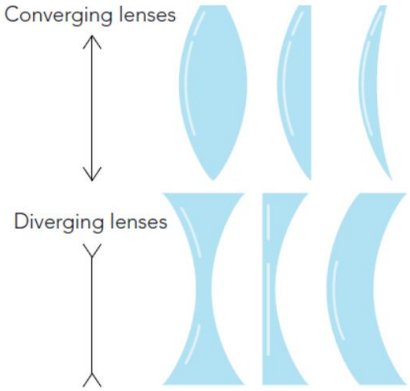
more diffraction: _____



less diffraction: _____



3.2.3 Thin lenses



_____（凸透镜/会聚透镜）

_____凹透镜/发散透镜）

➢ **Physical terms about the lens**

_____焦点）：_____

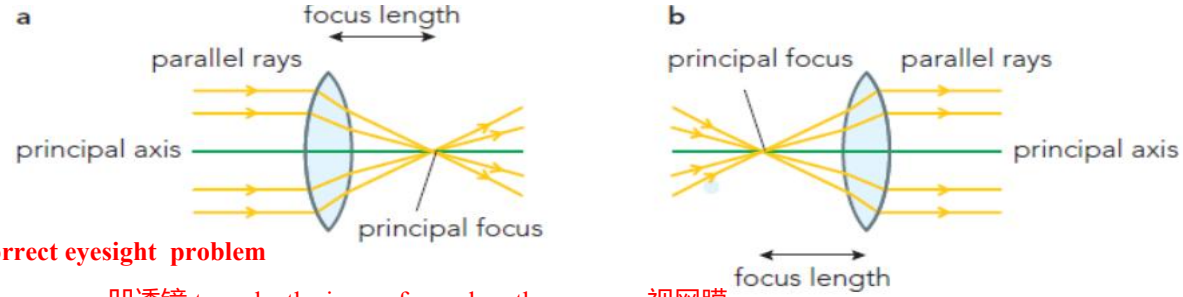
•_____焦距）：_____

•_____主光轴）：line that goes through optical center.

•_____光心）：the center of the lens

•_____实像）：image can be caught on a screen;when object is further away from the optical centre than **one F**

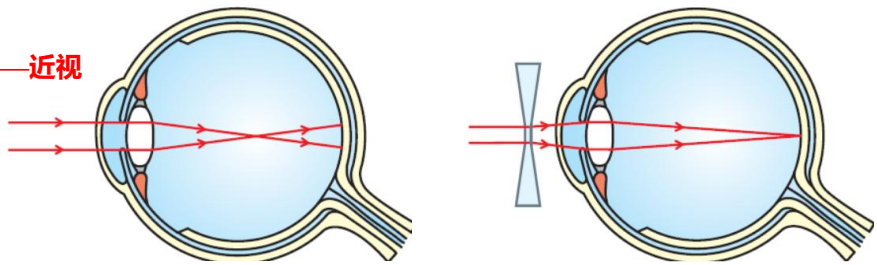
•_____虚像）：image cannot be caught on a screen When the object is closer to the optical centre than **one F**.



➢ **Using lense to correct eyesight problem**

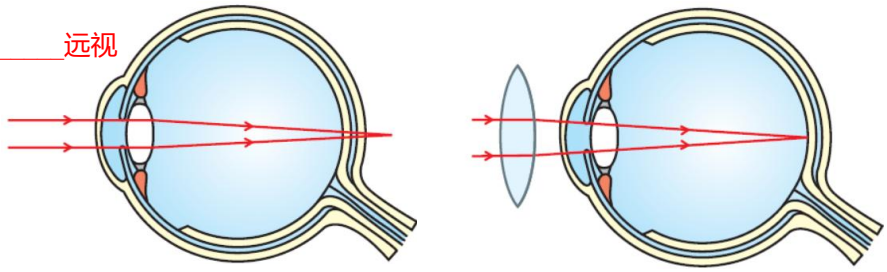
● ---Using _____凹透镜 to make the image formed on the _____视网膜.

_____近视



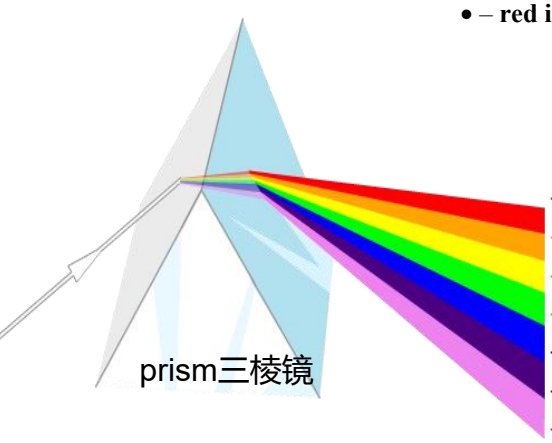
● ---Using_____凸透镜 to make the image formed on the _____视网膜.

_____远视



3.2.4 _____ of light光的色散

- **White light is a mixture of colors**, and the prism refracts each color by a different amount due to different speed of colourful light
- – **red is refracted least,violet most;**



– red-------- wavelength, speed , refracted
– orange
– yellow
– green
– blue
– indigo
– _____--------shortest _____, smallest _____, most _____

- _____单色光: _____单一频率的光 (used in laser激光)

3.3 _____电磁光谱

fλ=c=_____ in vacumn（真空）;
the greater _____,the smaller _____,

Radiation	Wavelength range / m
无线电波 _____	> _____
微波 _____	_____
红外线 _____	_____
可见光 _____	_____
紫外线 _____	_____
X-光波 _____	_____
γ-光波 _____	_____

Properties the E.M. waves（电磁波） have in common相同的:
i) they are all transverse wave
ii) they all can travel through vacuum
iii) they have same speed in vacuum
$c=3\times10^8$m/s
iv) they all can transfer energy and information

（1）.typical uses of the different regions of the electromagnetic spectrum including:

- **Radio waves（无线电波）**:_____广播 (TV, AM/FM radio) _____无线电频率识别
- **Micro waves（微波）**: _____手机通信, _____卫星通信, _____无线网络, _____微波炉加热
- **Infrared**: _____遥控器,intruder alarm, _____电烤炉, _____热成像and _____安保系统, _____ **光纤通信**

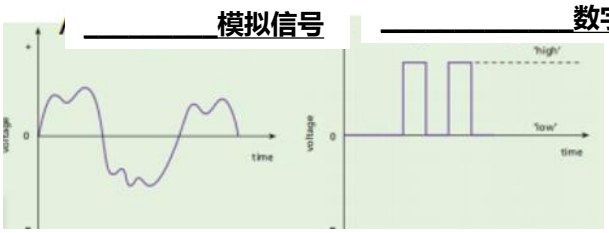
visible light: vision, photography, illumination

- **UV(ultra violet) rays（紫外光）**: _____**晒黑** and _____**灭菌** , security marking, detecting fake ,bank notes,
- **X-rays（X-射线）**_____**医学成像**& CT scan,
- **Gamma rays（r-ray）**: _____医疗灭菌,_____癌症治疗

通信信号有两种:

- Analogue uses a whole range of continuous variations to transmit a signal.
- Digital signals use only 2 states, on(high) and off(low).

_____**模拟信号** _____**数字信号**



3.4 Sound

(2) Speed of sound waves:wave speed depends on medium.

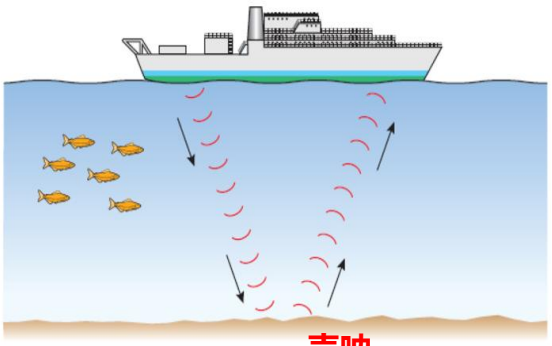
eg:Air : m/s ; Water: m/s ; Concrete : ; Steel: m/s
sound wave move in solid ,then liqiud , in gas.

(4) three properties of sound note:
音调) --increase with frequency
响度) -- incease with amplitude
音色) --determined by wave form
可听到频率）:Humans can hear frequencies between Hz.

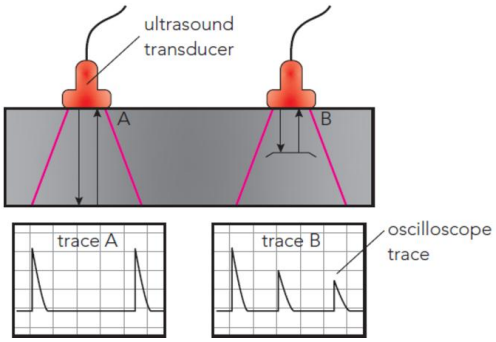
(5) 回声: The sound wave is reflected back from the surface.
use: 回声定位.

---Sound with frequency higher than 20kHz, is called 超声波）
---Sound with frequency lower than 20Hz, is called 次声波）

超声波应用



声呐



material testing



医学成像（B超）

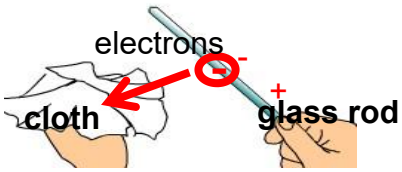
4 Electricity and Magnetism

4.2.1 Electric charge电荷

amount of charge: symble--**Q(q)** ;unit---**C(_____库伦)**

_____ (异种电荷吸引 and
_____ (同种电荷排斥)

(3) charging a body--the result of **loss or gain in electrons**(movement of electrons).
When you _____摩擦 a glass rod (or bar) with a cloth, electrons from the rod move onto the cloth. So **the cloth _____becomes negatively charged** overall, and the **rod _____ is positively charged**



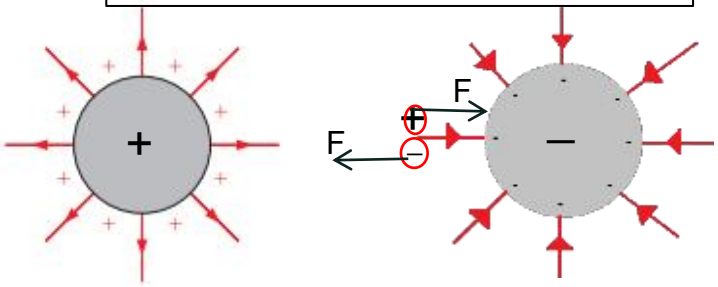
(4) and **conductors**（导体）:

_____绝缘体: A material through which **electrons _____** , that ***can*** be charged by friction.

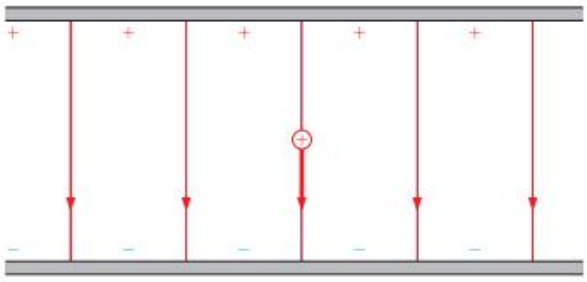
_____conductors导体: A material through which electrons_____.

(5)**_____电场):**_____
field lines indicate the **direction of**_____

field lines pattern for _____点电荷



field lines pattern between two charged parallel plates _____匀强电场



charged conducting sphere=**point charge** concentrated at its centre
inside the conducting sphere,there is no electrical field.

(6) define current:_____ **I=Q/t** **unit:_____安培**

(7) define E.M.F:_____ **unit:_____伏特， 1V=_____**

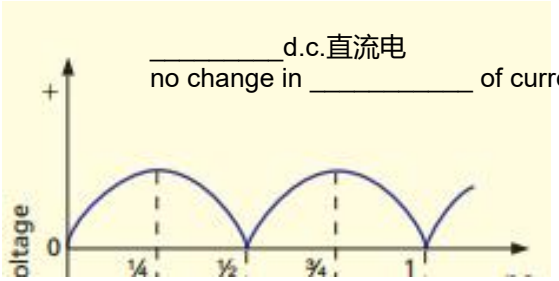
$$E.m.f(E) = \frac{W(\text{work done})}{Q} = \frac{\Delta E(\text{electrical energy supplied})}{Q}$$

(8) define voltage/Potential difference:_____

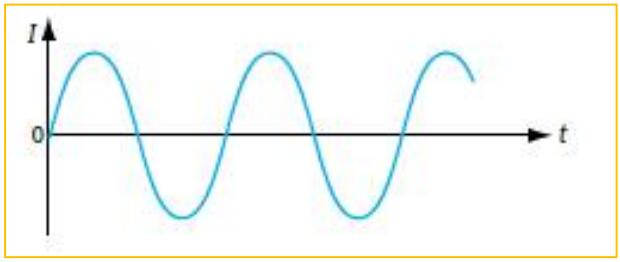
$$V = \frac{W(\text{work done})}{Q} = \frac{\Delta E(\text{electrical energy dissipated})}{Q}$$

(9) Resistance $R = \frac{V}{I}$ **unit:_____欧姆**

_____d.c.直流电
no change in _____ of current



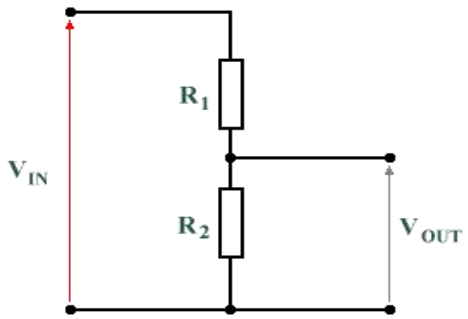
_____a.c.交流电
changed _____



千瓦时1kWh=1000Wx3600s=3.6X10^{6}J
(1h=1hour=3600s)

meaning:energy transferred in _____ at a rate of transfer of _____

use of potential divider circuit **串联分压电路的应用**



$V_1/V_2=R_1/R_2$

$V_{out}= V_{in}R_2/(R_1 +R_2)$

V_{out} depends on the ratio of R_1 and R_2

基本关系就是

_____电压和电阻成正比

电压大小取决于电阻R1与R2之比

4.4 Electrical safety

•**Damaged insulation（绝缘层破损）:** contact with the_____**火线** due to gap in the insulation causes _____**触电）** which can cause serious injury or shock.

precausion:

- _____**保险丝）:** A **fuse** protects a circuit. Thin piece of wire which_____It is placed on the _____火线 before the switch.

- _____**跳闸开关:** a safety switch that automatically open(trip) electricity from flowing when a current exceeds a certain value.
- **Benefits of _____接地） a Metal Case:**
  - Many electrical appliances, have metal cases, the earth wire creates a safe route for current to flow through, if live wire touches casing Earth terminal connected to metal casing, so the current goes through _____地线instead of causing an _____**触电**.

4 magnetism

4.1 Simple phenomena of magnetism

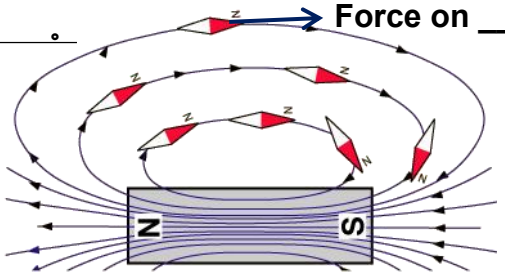
4.1 .1 magnetism

(1)**forces between magnets磁铁**

同极相斥 and 异极相吸

direction of field

(2) **磁场:** 。



(3) **磁性材料（ ）** ---such as i 铁, s 钢, n 镍 and c 钴

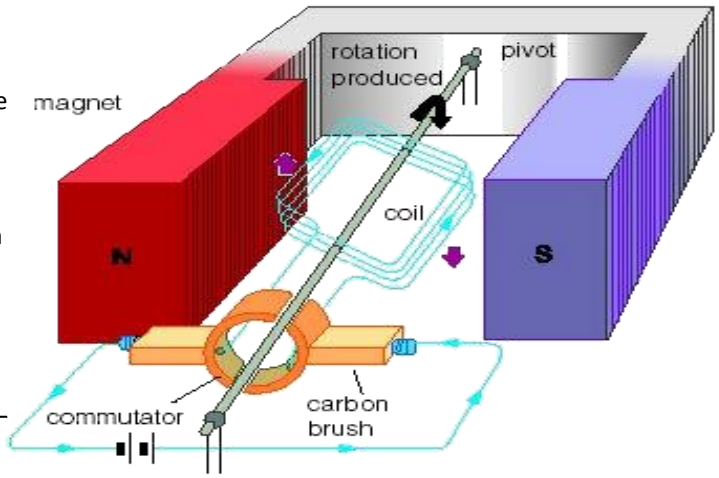
(5) Magnetic Properties of Iron and Steel

- **Iron铁** is **软磁铁--- temporary magnetism(**Gets magnetised faster but loses its magnetism as soon as inducing magnet is removed).
  use : 变压器铁芯 &电磁继电器 or 电磁铁
- **steel钢** is **硬磁铁---permanent magnetism(**Slow to be magnetized but retains magnetism for a long time) **use:making magnet**

(6) three ways to increase the strength of an **electromagnets电磁铁 or solenoids螺线管**

- _____
- _____
- _____ .

4.1.3 直流电动机



(1)how a DC motor runs on a direct current.

- When a current-carrying coil is in a magnetic field, magnet exerts force on current
- there are **two** _____ **with** _____**&**_____
  to produce _____

The coil is made of insulated copper wire and is free to _____between the poles of the magnet.

- The _____**开环换向器** is fixed to the coil and rotates with it.
- When the coil overshoots the vertical, **the commutator** ___ _____**, so the forces**_____**and**_____ _____.
- The **brushes are two** _____ **which rub against the commutator and keep the coil** _____ **to battery.**

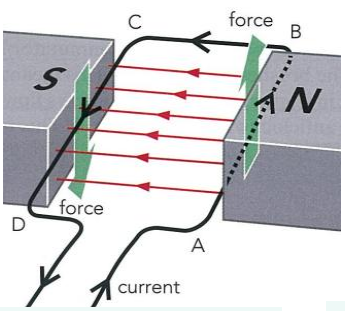
(2)

TURNING EFFECT INCREASED BY:

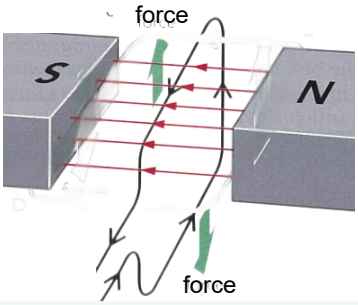
- _____
- _____
- _____

REVERSING ROTATION CAN BE DONE BY:

- _____
- _____



- **moment=**_____
- when the coil plane is_____to magnetic field.



- **moment=**_____

when the coil plane is at_____ to the magnetic field.

4.5 Electromagnetic effects

4.5.1 电磁感应 = _____磁场变化

Show understanding that when a conductor moving across a magnetic field then _____变化的磁场linking with a conductor can_____**感应电动势** in the conductor

(1) Describe an experiment to demonstrate electromagnetic induction

(2)State the factors affecting the **magnitude大小** of an **induced e.m.f感应电动势.**

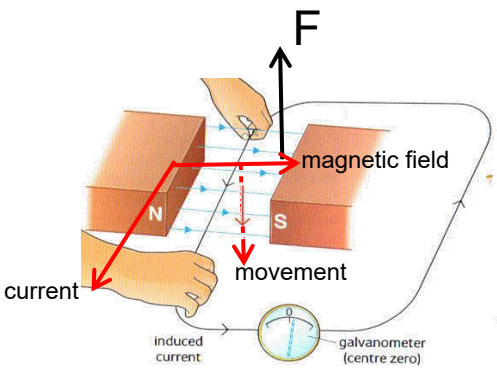
(i) _____

(ii)_____

(iii) _____

(3)Show understanding that the direction of an induced e.m.f (or current)_____

"the force on induced current alway "_____" the _____of magnet or coil

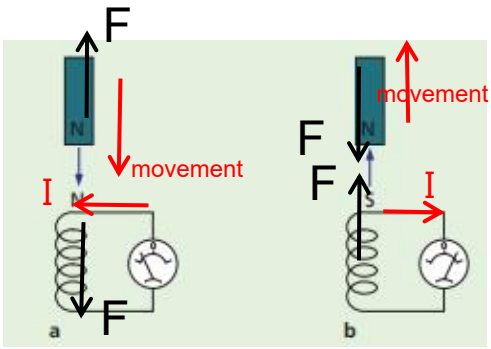


➢ example 1

wire"_____" the magnetic field downward

-------> _____

----------> _____ on induced current upward to _____ the motion downward



➢ example 2

- magnet push towards the coil

  ------> _____

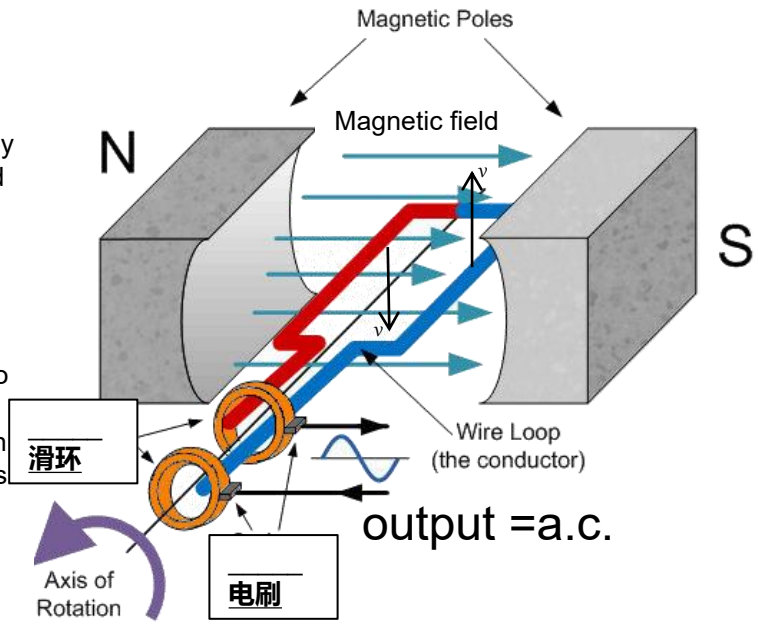
  ------->force on induced current to_____ the magnet moving close

- magnet push away the coil

  ------> current induced

  ------->force on induced current to _____ the magnet moving away

4.5.2 The a.c. generator 交流发电机



(1) how to work

- The coil is made of insulated copper wire and is _____by turning the shaft; the _____**滑环** are fixed to the coil and rotate with it.
- The _____**电刷** are two_____which rub against the _____ and keep the coil _____ the outside part of the circuit, usually made of carbon.
- **slip rings used _____**
- When the coil is rotated, it_____magnetic field lines, so an EMF is generated, which makes a current flow.
- Each side of the coil travels upwards then downwards then upwards etc. so the current flows backwards then forwards then backwards etc. so it is an alternating current.

(2)current change with time

- **induced current I = _____**
  when the coil plane is _____ to magnetic field,

- **induced current I=_____**
  when the coil plane is _____垂直 to the field, since it is cutting NO field lines.

(3)T**he induecd EMF can be increased by:**

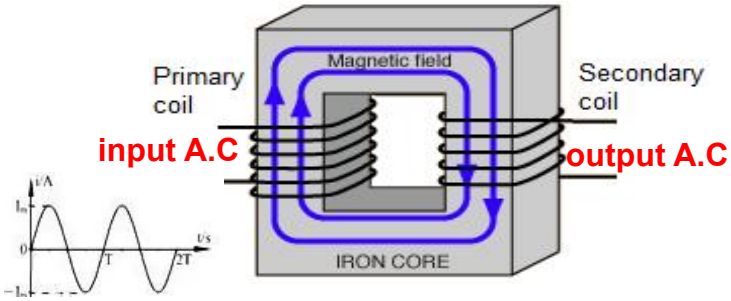
- **_____**
- **_____**
- **_____**

4.5.3 _____变压器

(1) how to work:

.

- _____
  _____
- _____
  _____
- _____
  _____.



- _____**升压变压器）** increases the voltage
- _____**降压变压器）** decreases voltage. for ideal transformer (NO energy loss)

_____ $=I_S/I_P$

input power= output power

.

$$I_p V_p = I_s V_s$$

the higherV ， the smaller I

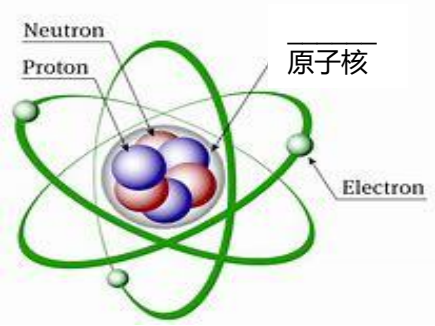
(2)the advantages of **high-voltage transmission高压输电**--- _____when the transmitted _____

explain: transmitted输送 current I=_____ **power lost in a resistor$P$ = _____**

having a lower current will _____ the power loss. so _____ voltage ,current get_____

IG physics Nuclear physics key Summary

1 .The nuclear model of the atom原子的核式结构



_____ 原子核

both_____质子 and_____中子are called _____**核子**

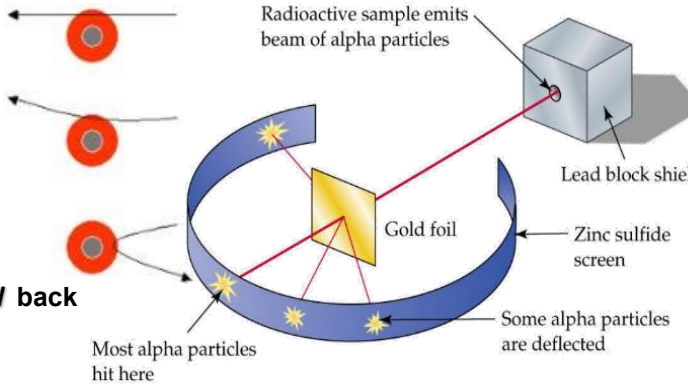
2.Conclusion of _____阿尔法粒子散射实验

- **Most *α particles are not deflected***

Show that：_____

A few α *particles are deflected*偏转 and
Extremely few (less than one in million) α *particles are deflected* back by nearly 180°

Show that：_____



3._____同位素: _____

4 _____核裂变：_____

- **_____核聚变：_____**

for all nuclear reaction, _____**and**_____ are conserved.

_____核能 released in the form of **γ ray**
due to there is total mass lost after reaction,
the total _____质能守恒.

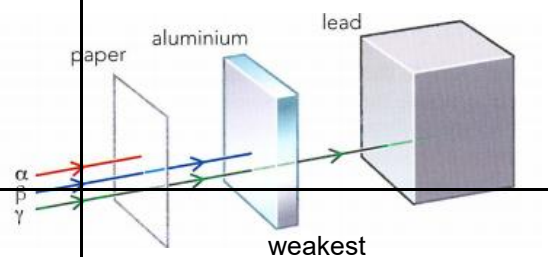
$^{1}_{1}H + {}^{2}_{1}H \rightarrow {}^{3}_{2}He$ + **nuclear energy（核能）γ**

total mass $^{1}_{1}H + {}^{2}_{1}H$ > total mass $^{3}_{2}He$

5._____背景辐射: _____

. It mainly comes from natural sources such (a) _____ (b) _____ (d)_____

- **characteristics（特点） of three kinds of radioactive放射性 emission射线**

	Alpha emission α 射线	**Beta β射线**	**Gamma γ ray γ 射线**
Nature	**Helium nucleus $_2$ He**	**electron**	**Electromagnetic Radiation**
Charge	**+ 2e**	**- e**	**No charge**
Mass	**4u** ^{4}	**1/1840 u**	**Massless**
_____穿透性	run a few centimetres in air stopped by a paper	run a few metres in air stopped by a thin aluminum	run untill stopped by a thick lead 
_____电离	most stongly (due to most +charge)	weak	weakest
deflection in magnetic and electric field电磁场中的偏转	deflected	deflected more (opposite to α)	no deflection
use	_____烟雾报警器利用电离性_____	**measure thickness due to the_____穿透性**	**(1)_____消毒;** **r(2)adiotherapy –γ rays kill_____** **(3)detect faults in manufactured goods**

6._____放射性衰变

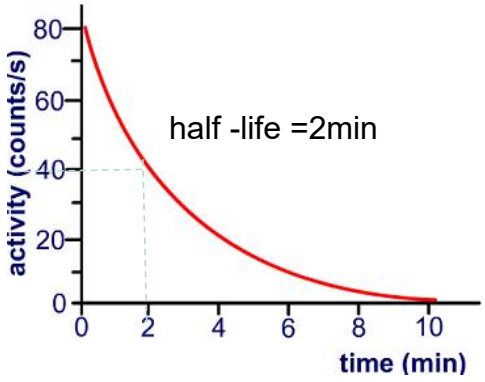
α-decay：equation: ${}^{A}_{Z}X \rightarrow$ ____ $+ \; {}^{4}_{2}\alpha$

β-decay：equation: ${}^{A}_{Z}X \rightarrow$ ____ $+ \; {}^{0}_{-1}e^{-}$

the electron come from inside the nucleus,one neutron has split into a proton and electron

${}^{1}_{0}n \rightarrow$ _____ + _____

7. _____半衰期:_____



IG-Space physics key summary

1 Earth,Sun and Moon

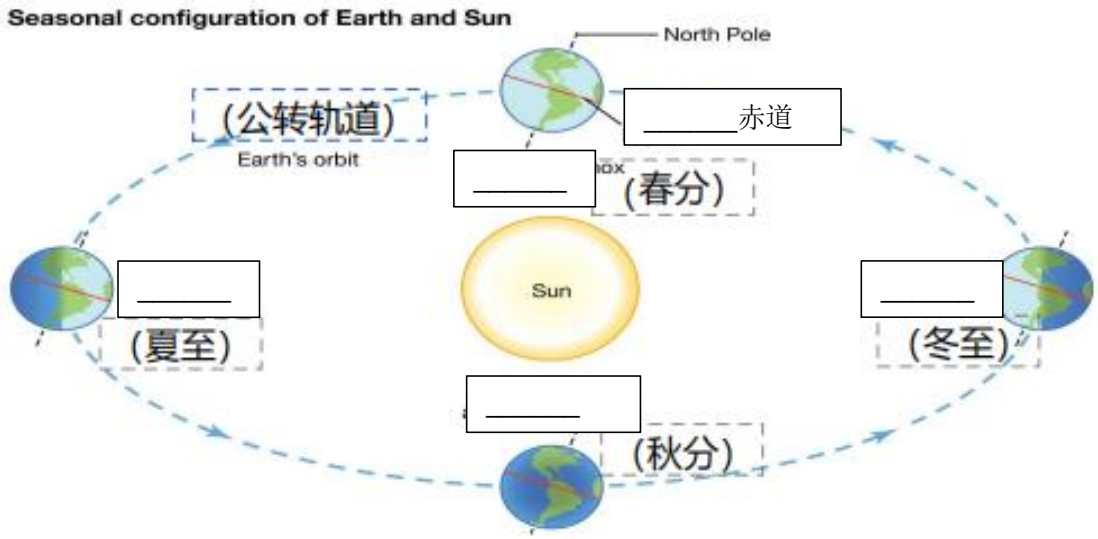
(1）.Explanation of day and night:

The Earth _____转动 on its _____地轴 ,It takes _____hours for Earth to complete one rotation.As the Earth spins on its axis,part of the Earth that faces the Sun has _____白天. And the part that faces away from the Sun has ___黑夜

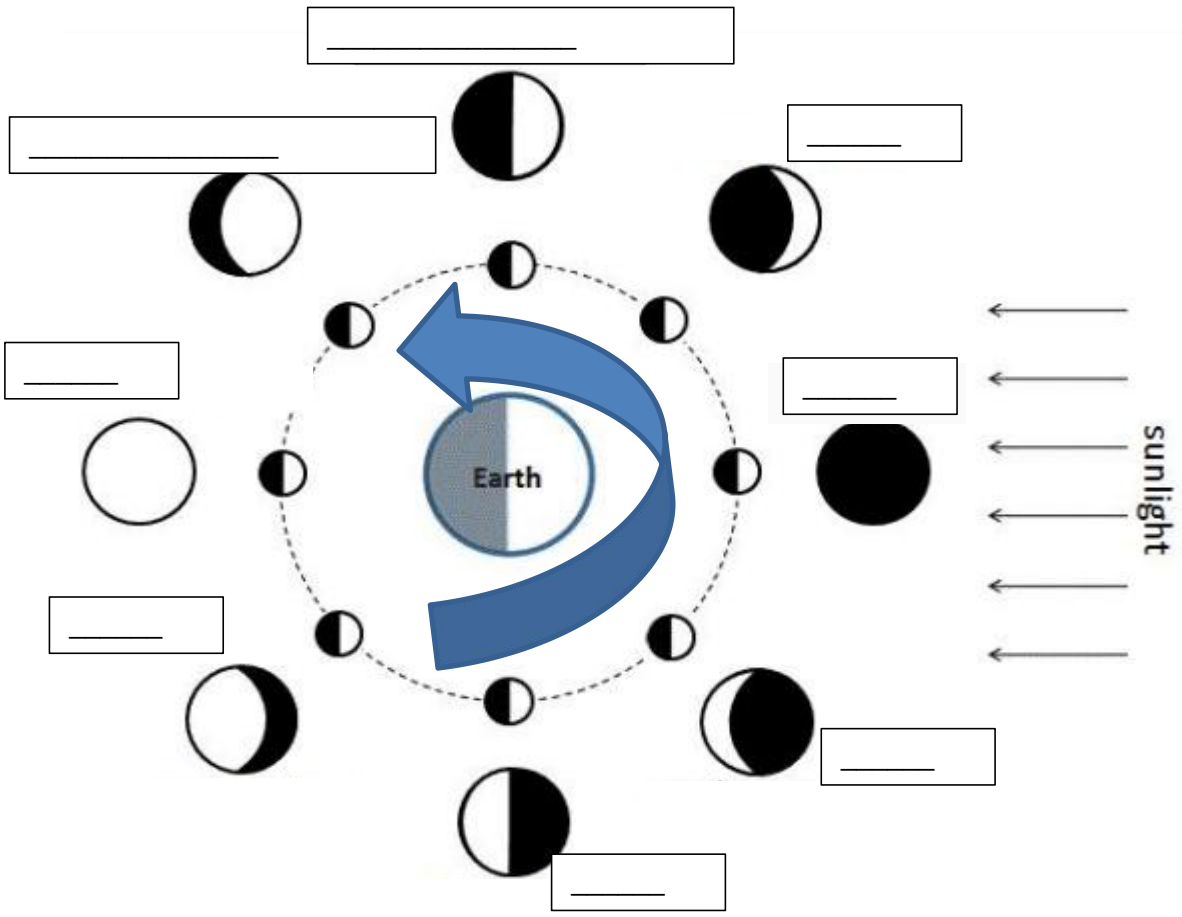
（2）.Explanation of the periodic nature of the seasons.

The Earth _____坏绕 the sun once_____一年. While it orbits the sun, it causes each _____半球 to either face the sun or face away from it.The_____半球 facing the sun has _____ and the _____ pointing _____it has winter.

• The _____地轴倾斜**causes seasons**.



（3)It takes approximately one month for the Moon to orbit the Earth and this is related to the periodic nature of the _____月相周期=27.5day



(4). The Solar System :

- one _____恒星： the Sun.
- the eight planets : 轨道半径越大，公转周期越_____,转速越_____(判断变大或者变小）

four _____岩质内行星(体积小，密度大）:_____（水星）_____（金星体表温度最高）, Earth, _____（火星）

four _____气态外行星（体积大，密度小）:_____(木星）, _____（土星卫星最多）, _____天王星 and _____海王星.

- minor planets that orbit the Sun,an object which orbits the Sun but is not large enough or far enough from another object,that include the_____矮行星 such as _____冥王星.

●moons:that orbit the planets. is the_____天然卫星

●Smaller Solar System bodies, including

_____小行星 in the_____小行星带 and_____流星体: rocks that are smaller than the planet which orbit the Sun

_____彗星: aballofice, dust and gas which orbits the Sun in a highly _____ 椭圆轨道.

_____吸积盘: a rotating disc of matter formed by_____沉积物: ____ and _____pulled together by gravity

(5) **.Gravitational field(引力场）：**

. The force that keeps an object in orbit around the Sun is the _____万有引力of the Sun.

- the strength of the gravitational field depends on the _____ and _____

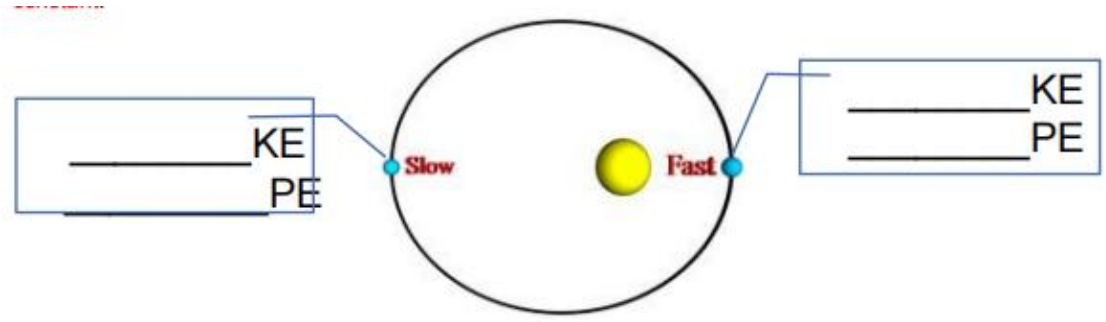
The strength of the Sun's gravitational field _____ and that the orbital speeds of the planets _____as the distance from the Sun _____(判断递增还是递减关系）.

理想圆轨道：orbital speed v = 2πr/T

r : _____环绕半径 T : _____环绕周期

实际_____椭圆轨道: moving fastest at closest to the Sun ,距离越大，速度越小

总能量total energy =KE+PE 不变



2. Stars and the Universe

- The Sun is a star in the_____**星系** known as the _____**银河系.**
- TThe Sun is a star of medium size, consisting mostly of **hydrogen**(_____%) and **helium**(_____%).
  the sun radiates most of its energy in the_____(50%), _____(40%)and _____(10%)
- Stable stars like our sun are powered by the _____**核聚变**. This makes the Sun shine.
- _____光年：_____

Describe the life cycle of a star:恒星的生命周期

(a) a star is formed from_____星际 clouds of _____ and _____ that contain **hydrogen**

(b) a _____原恒星is on interstellar cloud collapsing and increasing in temperature as a result of its internal _____万有引力。

(c) a _____原恒星 becomes a _____ when the _____ is balanced by an _____force due to the high temperature in the centre of the star

(d) all stars eventually run out of _____ as f_____ for the nuclear reaction

根据恒星质量不同演化两条路径

- For more massive stars

more massive stars expand to form _____红超巨星 when most of the hydrogen in the centre of the star has been corverted to helium，then explodes as a _____超新星, forming a _____星云 containing hydrogen and new heavier elements,leaving behind a **neutron star**（中子星）or a **black hole**（黑洞）

- For less massive star(such as SUN)

a ____红巨星from a less massive star forms a _____行星状星云 with a _____白矮星at its centre.

表格

英文术语	中文译名	关键考点
Nebula	星云	恒星诞生的摇篮，主要成分氢和氦
Protostar	原恒星	引力坍缩阶段，未启动核聚变
Stable star		
Main-sequence star	主序星	氢聚变稳定进行，恒星最长的阶段
Red giant/Red supergiant	红巨星 / 红超巨星	氢耗尽后膨胀的阶段，由质量决定类型
Planetary nebula	行星状星云	中小质量恒星抛射的外层气体
White dwarf	白矮星	中小质量恒星的核心残骸，电子简并支撑
Supernova	超新星	大质量恒星的剧烈爆炸，抛射重元素
Neutron star	中子星	大质量恒星核心残骸，中子简并支撑
Black hole	黑洞	超大质量恒星核心坍缩的产物，引力极强

The universe

- _____红移 as an _____in the observed _____ of electromagnetic radiation emitted from _____远离的stars and galaxies.

- The _____红移 in the light from distant galaxies is evidence that the universe is _____膨胀and supports the_____（宇宙大爆炸理论）

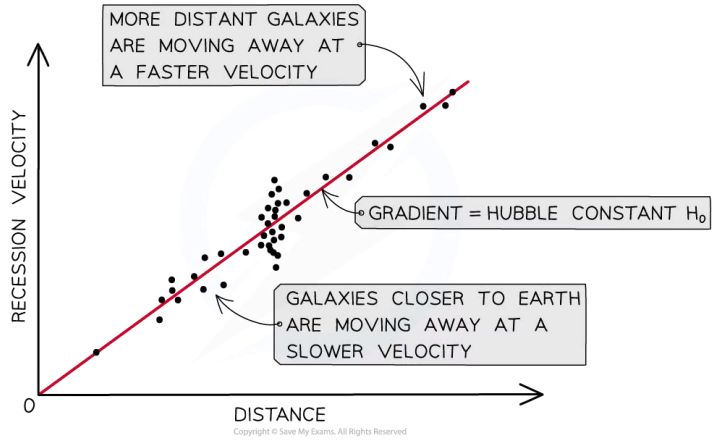
- The _____ of a specific frequeny is observed at all points in space Around us and is known as _____(宇宙微波背景辐射)(CMBR)

The CMBR was produced shortly after the _____ and that this radiation has been expanded into the_____ region of the electromognetic spectrum as the Universe _____. And that this is evidence for the idea that all the matter in the Universe was present at a _____.

- Hubble's law shows that _____

$$H_0 = \frac{v}{d}$$

v: _____
d: _____
H_0: _____



The equation $\frac{1}{H_0} = \frac{d}{v}$ represents an estimate for the _____ of the Universe.